MILITARY STANDARD

FASTENER SYSTEMS FOR AEROSPACE APPLICATIONS

TO ALL HOLDERS OF MIL-STD-1515A:

1. THE FOLLOWING ITEMS OF MIL-STD-1515A HAVE BEEN REVISED AND SUPERSEDE THE ITEMS LISTED:

NEW ITEM	DATE	SUPERSEDED ITEM	DATE	TYPE CHANGE
Page iii	24 Sept 1979	REPRINTED WITHOUT CHANGE		
Page iv	24 June 1983	Page iv	5 June 1981	Minor
Page v	24 June 1983	Page v	5 June 1981	Minor
Page vi	12 July 1978	REPRINTED WITHOUT CHANGE		
Page 3	24 June 1983	Page 3	24 Sept 1979	Typo
Page 4	24 June 1983	Page 4	24 Sept 1979	Minor
Page 103.7	24 Sept 1979	REPRINTED WITHOUT CHANGE		
Page 103.8	24 June 1983	Page 103.8	24 Sept 1979	Minor
Page 112.1	24 June 1983	Page 112.1	24 Sept 1979	Minor
Page 112.2	24 Sept 1979	REPRINTED WITHOUT CHANGE		
Req 120	24 June 1983	INITIAL PUBLICATION		
Req 201	24 June 1983	INITIAL PUBLICATION		
Req 202	24 June 1983	INITIAL PUBLICATION		
Page 204.3	24 Sept 1979	REPRINTED WITHOUT CHANGE		
Page 204.4	24 June 1983	Page 204.4	24 Sept 1979	Minor
Page 204.13	24 Sept 1979	REPRINTED WITHOUT CHANGE		
Page 204.14	24 June 1983	Page 204.14	24 Sept 1979	Typo
Req 206	24 June 1983	Req 206	5 June 1981	Major
Req 213	24 June 1983	INITIAL PUBLICATION		

2. Change 2 renumbered Requirement 215 as Requirement 129 but inadvertently failed to delete Requirement 215. This change deletes Requirement 215. Remove Requirement 215 from the Standard.

3. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

4. Holders of MIL-STD-1515A will verify that page changes, deletions and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or cancelled.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing Activity:
Air Force - 11

Project 53GP-0039

Review Activities:
Army - AR, MI
Navy -
Air Force - 10, 99
DLA - IS

FSC 53GP

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TABLE 201-I. Non-locking screws and bolts - continued

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC	FINISH 1/	OTHER INFO
160 KSI FTU Tension - continued						
100° Flush	Hi-Torque	Steel	NAS1221	NAS498	Cadmium	sizes .1120-40
100° Flush	Hi-Torque	A286	NAS1221E	AMS7478/ AMS7479	Passivated	Thru .1640-32, for larger sizes use NAS1580
100° Flush	Hi-Torque	Ti	NAS1221V	NAS621	Bare	
100° Flush	Hi-Torque-Torq-Set	Steel	NAS1580A	NAS498	Cadmium	
100° Flush	Hi-Torque-Torq-Set	A286	NAS1580C	AMS7479	Passivated/ Cadmium	
100° Flush	Hi-Torque-Torq-Set	Ti	NAS1580V	NAS621	Bare	
100° Flush	Hi-Torque	Steel	NAS1219	NAS498	Cadmium/ Aluminum	Full Thread
100° Flush	Hi-Torque	A286	NAS1219E	AMS7479	Passivated/ Cad/Alum	Full Thread
100° Flush	Hi-Torque	Ti	NAS1219V	NAS621	Bare/ Aluminum	Full Thread
100° Flush	Torq-Set	Steel	NAS1102	NAS498	Cadmium	Full Thread
100o Flash	Torq-Set	A286	NAS1102V	AMS7478/ AMS7479	Passivated	Full Thread
100o Flush	Torq-Set	Ti	NAS1102V	NAS621		Full Thread
180 KSI FTU Tension						
Protruding	12 Spline	Steel	MS21134	MIL-B-8831	Cadmium	
Protruding	12 Spline	Steel	MS14157	MIL-B-8831	Cadmium	Wheel Bolt
Protruding	Hex	Steel	NAS1953-1970	NAS4002	Cadmium	
Protruding	Hex	A286	NAS1953C-1970C	NAS4003	Passivated/ Aluminum	
Brazier	Torq-Set/ Hi-Torque	Steel	NAS1982-1990	NAS4002	Cadmium	No Fatigue Requirement
Brazier	Torq-Set/ Hi-Torque	A286	NAS1982C-1900C	NAS4003	Passivated	

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TABLE 201-I. Non-locking screws and bolts.

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC	FINISH 1/	OTHER INFO
160 KSI FTU Tension						
Protruding	Hex	Steel	NAS6603-6620	NAS4002	Cadmium	Long Thread
Protruding	Hex	A286	NAS6703-6720	NAS4003	Passivated	Long Thread
Protruding	Hex	Ti	NAS6803-6820	NAS4004	Bare	Long Thread
Protruding	Hex	Steel	NAS6203-6220	NAS4002	Cadmium	Med. Thread
Protruding	Hex	A286	NAS6303-6320	NAS4003	Passivated	Med. Thread
Protruding	Hex	Ti	NAS6403-6420	NAS4004	Bare	Med. Thread
Protruding	Hex	Steel	NAS563-572	MIL-B-6812	Cadmium	Full Thread
Protruding	Hex	A286	NAS563-572C	MIL-B-6812	Passivated	Full Thread
Pan	Hi-Torque	Steel	NAS1218	NAS498	Cadmium	Sizes-1120-40 thru .1640-32 for larger sizes use NAS1578
Pan	Hi-Torque	A286	NAS1218E	AMS7478/ AMS7479	Passivated	
Pan	Hi-Torque	Ti	NAS1218V	NAS621	Bare	
Pan	Torq-Set	Steel	NAS1578A	NAS498	Cadmium	
Pan	&	A286	NAS1578C	AMS7479	Passivated	
Pan	Hi-Torque	Ti	NAS1578V	NAS621	Bare	
Pan	Hi-Torque	Steel	NAS1216	NAS498	Cadmium	Full Thread
Pan	Hi-Torque	A286	NAS1216E	AMS7479	Passivated	Full Thread
Pan	Hi-Torque	Ti	NAS1216V	NAS621	Bare	Full Thread
Pan	Torq-Set	Steel	NAS1101	NAS498	Cadmium	
Pan	Torq-Set	A286	NAS1101E	AMS7478/ AMS7479	Passivated	Drilled Head Only
Pan	Torq-Set	Ti	NAS1101V	NAS621	Bare	Approved

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FOREWORD

The purpose of this document is to establish standardization in the selection, development, and use of aerospace fastening systems, including hole characteristics and inspection criteria. Existing fasteners cover a wide variety of configurations, sizes, materials, and finishes. Various aerospace fastener systems have common fastener needs but lack defined application exchangeability. This document provides contractual design requirements and guidelines for proper selection and application of approved fastening systems. The basic objectives of this standard can be summarized as:

> To document selection and application information for approved fastening systems and procedures so the designer can achieve an optimum balance of performance, reliability, and exchangeability with minimum cost, logistic inventory and maintenance.

Selections and procedures are limited to those listed herein.

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	131 *		Fastener Temperature Limitations
	132 *		Mechanical Fastening in Composites

*Denotes change
Supersedes page iv of 5 June 1981

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SECTION 400 * PROPULSION SYSTEMS FASTENER REQUIREMENTS (PROPOSED)

SECTION 500 * HYDRAULIC SYSTEMS FASTENER REQUIREMENTS (PROPOSED)

SECTION 600 * ELECTRICAL/ELECTRONIC FASTENER REQUIREMENTS (PROPOSED)

SECTION 700 * SUBSYSTEMS/AUXILIARY FASTENER REQUIREMENTS (PROPOSED)

SECTION 800 * APPLICATION REQUIREMENTS (PROPOSED)

* Denotes change.
Supersedes page v of 12 July 1978

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3.1.11 Fastening system. An installed fastener, its component parts, the geometry of the hole where it affects the performance of the system, and installation and removal tooling and procedures.

3.1.12 Grip or grip accommodation. The allowable variation of material thickness in which a specific fastener can function.

3.1.13 Impedance locking. A resistance that impedes disengagement of mating fastener components.

3.1.14 Locking device. A part or mechanism designed to prevent loss of preload or disengagement of a fastener.

3.1.15 Permanent fastener. A fastener which does not require removal to service or maintain the aircraft and that normally requires the destruction of the fastener or one of its components during removal.

3.1.16 Plate nut. (See anchor nut)

3.1.17 Self locking. An attribute of a fastener or fastener assembly having an integral locking element to impede relative rotation of mating components.

* 3.1.18 Shank. The unthreaded portion of a fastener between the head and the threads, locking grooves, formed head, etcetera.

4. GENERAL STATEMENTS

4.1 Guidelines. A separate publication, "AMFRG Manual on Organization and Operations of the Aeromechanical Fastener Requirements Group for MIL-STD-1515", has been distributed to organizations participating in the preparation of MIL-STD-1515. This publication presents guidelines for the preparation and coordination of requirements for inclusion in this standard. These guidelines shall be followed for all requirement additions or changes and shall be revised as necessary by the preparing activity or his designee.

4.2 Application

a. The sections contained herein are intended to provide uniform requirements applicable to mechanical aerospace fasteners and shall be incorporated by reference in general and detail weapon systems and equipment specifications.

b. If a requirement contained herein conflicts with a requirement in the general or detail weapon system or equipment specification, the weapon system or equipment specification shall take precedence. If a requirement listed in the contents has not been published but is referenced herein, it is not applicable and the requirements of the general or detail weapon system or equipment specification shall apply.

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c. Section 100 Requirements apply only to Section 200 Structural fastener requirements (excluding subsequent proposed sections such as nonstructural, propulsion, etcetera, not published or in preparation).

4.3 Descriptive factors and symbols. The descriptive factor symbols used in this standard shall be in accordance with MS17855.

5. NOTES

5.1 Availability of Documents. Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

6. INTENDED USE AND GUIDANCE CRITERIA

6.1 Information and guidance. Paragraph 6 in all applicable requirements in this document presents information and guidance for use of that requirement or parts listed therein. The paragraph is not contractual.

6.2 Implementation. This standard may not be applicable to programs in the experimental, developmental, exploration, prototype, demonstration, validation program phases where the advantages of the standard will not be realized.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing Activity:
Air Force - 11

Reviewer Activities:
Army - AR, MI
Navy -
Air Force - 10, 99
DLA - IS

Project No. 5300-0039

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RELEASE FOR USE OF NONAPPROVED FASTENERS

1. Purpose. Fastening systems, fasteners, and installation requirements identified and listed in the applicable requirements of this standard are approved and shall be given selection priority in new aerospace systems design. Those not listed as approved for use require a release for use by the procuring activity prior to incorporation into design, during development, or for use in assembly during production. This requirement establishes preparation and submission procedures for justification and engineering data necessary for the evaluation and use of fasteners not listed herein. It also establishes the point of contact for release requests.

2. Documents applicable to Requirement 112

MIL-D-8706	Data and Tests, Engineering, Contract Requirements for Aircraft Weapon Systems
MIL-STD-965	Parts Control System

3. Procedures for use. When a determination has been made that a fastening system or fasteners not listed herein should be used, a request for release for use, with the justification and engineering data specified herein, shall be submitted to the procuring activity in accordance with the procedures of MIL-D-8706 and MIL-STD-965. When contracts do not call out any of the above documented procedures, the requests shall be made to the contact points listed in table 112-1, through the procuring activity in accordance with procedure 1 of MIL-STD-965 and the data requirements of 3.1 of this requirement.

3.1 Engineering data required. When applicable, the following engineering data shall be furnished for each fastener not listed herein for which release for use is requested.

a. The reason an approved fastener would not be a satisfactory selection to perform the function required by the design problem or application.

b. The specific materials from which the fastener is manufactured.

c. The specific plating, coatings, surface treatments, and lubricants, as applicable. These shall be completely identified.

d. Values for the physical and mechanical properties, and available installation performance requirements, and available structural design load allowable data.

* e. An illustration of the fastener with the dimensions required to establish the geometrical material limits necessary for design selection and for determining exchangeability. This includes the geometrical material limits of installation formed fasteners after installation and clearances required for installation.

f. The maximum weight limit per one hundred of the fasteners.

g. The design and usage limitations to be applied.

h. The installation or process instruction for the part.

* Denotes change.

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3.2 Contract data requirements. Contract data requirements shall include the data specified in 3.1.

TABLE 112-I. Request contact points

Department	Submit to	Action Required
Air Force	Aeronautical Systems Division ATTN: ENFEM Wright-Patterson AFB, OH 45433 Telephone: 513-255-4158	Disapprove or Release
Army	US Army Aviation R&D Command ATTN: DRDAV-EKS P. O. Box 209 - Main Station St. Louis, MO 63166 Telephone: 314-268-5791	Disapprove or Release
Navy	Naval Air Development Center ATTN: 6013 Warminster, PA 18974 Telephone: 215-441-2833	Review and Recommendation to NAVAIR
	Naval Air Systems Command ATTN: AIR-5303 Washington, DC 20361	Disapprove or Release
All Military Services DISC/MPCAG	Defense Industrial Supply Center ATTN: DISC-EPM Philadelphia, PA 19111 Telephone: 215-697-4395	Information and Recommendation

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DRIVE AND WRENCHING ELEMENTS

1. Purpose. This requirement provides for the selection of drive and wrenching elements established as approved for use in aerospace structure fastener systems. Selection may be further limited by other documents.

2. Documents applicable to Requirement 120

2.1 Government documents

Federal Specifications

GGG-W-636	Wrenches, (Box, Open End, and Combination)
GGG-W-660	Socket, Socket Wrench and Attachments (For Power Driven Impact Wrenches)
GGG-W-1437	Wrench, Socket and Box End (Thin Wall-High Strength)

Military Specifications

MIL-W-8982	Wrenches, Splined, Socket and Box, Square Drive, High Strength, Thin Wall
MIL-B-9946	Bits, Screwdriver, General Specification for

Military Standards

MS 9006	Recesses-Cross, Low Torque Drive, Dimensions and Gage Dimensions for
MS 14191	Recess, Ribbed-Torq-Set, Dimensions of Recess, Gage and Driver for
MS 21132	Wrenching Element, External Hexagon, for Threaded Fasteners
MS 33750	Recess, Hi-Torque; Dimensions of Recess, Gage, and Driver for
MS 33781	Recess - Torq-Set, Dimensions of Recess, Gage, and Driver for
MS 33787	Wrenching Element, External Spline, Dimensions for

Other Government Documents

TO-1-1A-8	Structural Hardware Section III

2.2 Non-Government documents

ANSI B18.3	Socket Cap, Shoulder and Set Screws - Inch Series (Including Dimensions of Hexagon and Spline Sockets and Keys to Match)
NAS 7100	Recess, Phillips®, Dimensions of Recess and Gages
NAS 7101	Bit, Screwdriver, Phillips®, Specification for

3. Design and usage limitations

3.1 External wrenching elements (external or internal threaded)

3.1.1 The hexagon drive external wrenching element in accordance with MS 21132 shall be limited to use for fasteners up to 180,000 psi maximum ultimate tensile stress only.

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3.1.2 All externally threaded fasteners heat treated to 180,000 psi minimum ultimate tensile stress and higher and all internally threaded fasteners rated at 180,000 psi minimum ultimate tensile stress and higher shall have the MS 33787 external wrenching element. Fasteners less than 180,000 psi may utilize the MS 33787 wrenching element.

3.2 Internal drive elements

3.2.1 The MS 9006 recess shall be limited to use in fasteners heat treated up to 150,000 psi maximum ultimate tensile stress.

3.2.2 The NAS 7100 recess may be used in fasteners heat treated up to 160,000 psi maximum ultimate tensile stress. (Note: Fasteners heat treated above 160,000 psi maximum ultimate tensile stress with MS 9006 or NAS 7100 recess may be used in secondary structure with acquisition activity approval.)

3.2.3 Recesses in accordance with MS 33750, MS 33781 and MS 14191 shall be used in fasteners heat treated to 160,000 psi minimum ultimate tensile stress and higher. These recesses may be used in fasteners below 160,000 psi in order to reduce the recess types used in each system.

3.2.4 Internal hexagon recess in accordance with ANSI B18.3 shall be limited to the threaded end of pins without head driving recess, set screws, and certain panel fasteners as specified in Requirement 129. Acquisition activity approval is required for use of socket head cap screws in airframe structures.

4. Design selection and approved callout

Drive and wrench elements identified by this requirement shall conform to the documents listed in section 2 and shall be selected from table 120-I.

5. Tooling

5.1 Bits for installing internal drive fasteners shall be in accordance with MIL-B-9946, MS 33781, MS 14191, MS 33750, ANSI B18.3, and NAS 7101, as applicable.

5.1.1 Refer to TO-1-1A-8, Section III for driver wear limits.

5.2 Wrenches for installing fasteners with spline or hexagon external wrenching elements shall be in accordance with MIL-W-8982, GGG-W-1437, GGG-W-636, as applicable. MIL-W-8982 wrenches are compatible with both the 12-spline and MS 21132 hexagon wrenching elements. Wrenches in accordance with GGG-W-636, GGG-W-660, and GGG-W-1437 are not compatible with the 12-spline wrenching element in accordance with MS 33787.

5.3 Recesses in accordance with MS 9006 and NAS 7100 and their related tools are fully interchangeable. Recesses in accordance with MS 14191 and MS 33781 and their related tooling are fully interchangeable.

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5.4 Care shall be exercised to ensure that the correct size bit or wrench is used and particularly for internal drives that specified installation torque requirements are not exceeded.

6. Intended use and guidance criteria

6.1 Some "off-the-shelf" aerospace systems may contain fasteners with drive and wrenching systems not specified herein. Field replacement with preferred drive and wrenching elements shall be dependent upon availability of equivalent hardware meeting all other required features (configurations, material, performance, et cetera) of the original fasteners.

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TABLE 120-I. Drive or wrench element selection.

DRIVE OR WRENCH ELEMENT	DESIGN STANDARD	TOOLING	DESIGN STRENGTH AND USAGE LIMITATIONS
Hexagon (External) 	MS 21132	GGG-W-1437 GGG-W-636 GGG-W-660	Less than 180,000 psi tensile stress. .190 inch nominal diameter and larger.
12 Spline (External) 	MS 33787	MIL-W-8982	180,000 psi minimum tensile stress. (Optional below this level)
Phillips (Internal) 	MS 9006	MIL-B-9946	150,000 psi maximum tensile stress.
	NAS 7100 (Ribbed)	NAS 7101	160,000 psi maximum tensile stress.
Torq-Set (Internal) 	MS 14191 (Ribbed)	MIL-B-9946	160,000 psi minimum tensile stress. (Optional below this level)
	MS 33781		
Hi-Torque (Internal) 	MS 33750	MIL-B-9946	160,000 psi minimum tensile stress. (Optional below this level)
Hexagon (Internal) 	ANSI B18.3	ANSI B18.3	See 3.2.4 for limitations.

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BOLTS AND SCREWS, REGULAR AND SELF-LOCKING

1. Purpose. This requirement provides for the selection of bolts established as approved hardware for use in aerospace structural joining.

2. Documents applicable to Requirement 201

2.1 Government documents

Federal Specification

GGG-W-1437	Wrench, Socket and Box End (Thin Wall-High Strength)

Military Specifications

MIL-B-6812	Bolts, Aircraft
MIL-B-7838	Bolt, Internal Wrenching, 160 KSI Ftu
MIL-B-8831	Bolt, 180 KSI Ftu and 108 KSI Fsu, 450°F Protruding and Flush Head; General Specifications for
MIL-B-8906	Bolt, Tensile, Steel, 220 KSI Ftu, 450°F External Wrenching, Flanged Head
MIL-F-8906/2	Bolts, Shear, Steel, 132 KSI Fsu, 450°F
MIL-B-8907	Bolt, Tensile, Steel, 260 KSI Ftu, 450°F External Wrenching, Flanged Head
MIL-F-8961	Fastener, Externally Threaded, 450°F and 1200°F, Self-Locking Elements for
MIL-W-8982	Wrenches, Splined, Socket and Box, Square Drive, High Strength, Thin Wall
MIL-B-9946	Bits, Screwdriver, General Specification for
MIL-F-18240	Fastener, Externally Threaded, 250°F Self-Locking Element for
MIL-B-87114	Bolts, Recess Drive, General Specification for

Military Standards

MS 14157	Bolt, Wheel, Tension, Flanged Steel, 180 KSI Ftu, 450°F, External Spline Drive
MS 14163	Bolt, Wheel, Tension, Flanged, Steel 220 KSI, Ftu, 450°F, External Wrenching, Spline, Drive
MS 14179	Nut, Plate, Self-Locking, Floating, Two Lug, Reduced Rivet Spacing, Steel, (Vespel Insert) 500 Cycles Reuse, Replaceable Nut, 160 KSI Ftu, 450°F
MS 14181	Bolt, Tension, Inconel, 220 KSI Ftu, 800°F, External Wrenching, Spline Drive, Flanged Head
MS 14191	Recess, Ribbed-Torq-Set, Dimensions of Recess, Gage and Driver for
MS 21134	Bolt, Tension, Steel, 180 KSI Ftu, 450°F, External Wrenching, Spline Drive, Flanged
MS 21296	Bolt, Tension, Steel, 260 KSI, Ftu, 450°F, External Wrenching, Spline Drive, Flanged Head
MS 21297	Bolt, Tension, Steel, 220 KSI, Ftu, 450°F, External Wrenching, Spline Drive, Flanged Head
MS 33750	Recess, Hi-Torque; Dimensions of Recess, Gage and Driver For
MS 33781	Recess-Torq-Set, Dimensions of Recess, Gage and Driver for

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Marshall Space Flight Center Documents

MSFC 10M90001 Requirements and Procedures for Contamination Control Due to Vacuum Outgassing

(Application for copies should be addressed to the Marshall Space Flight Center, Huntsville, AL 35812)

2.2 Non-Government documents

NAS 498	Bolts, Shear
NAS 563-572	Bolt, Full Threaded, Fully Identified Head
NAS 621	Fastener, Titanium Alloy, Procurement Specification
NAS 1101	Screw, Machine, Flat Fillister Head, Full Thread, Torq-Set
NAS 1102	Screw, Machine Flat 100 Deg Head, Full Thread, Torq-Set
NAS 1161-1168	Screw, Self-Locking, Flat 100 Deg Head, Shear, Torq-Set
NAS 1189	Screw, Self-Locking, Flat 100 Deg Head, Full Thread
NAS 1190	Screw, Self-Locking, Pan Head, Full Thread
NAS 1216	Bolt, Flat, Pan Head, Hi-Torque Recess, Full Thread
NAS 1218	Bolt, Flat, Pan Head, Hi-Torque Recess, Long Thread
NAS 1219	Bolt, 100° Flush Tension Head, Hi-Torque Recess, Full Thread
NAS 1221	Bolt, 100° Flush Tension Head, Hi-Torque Recess, Long Thread
NAS 1578	Bolt, Flat Pan Head
NAS 1580	Bolt, 100 Deg Flush Tension Head
NAS 1581	Bolt, 100 Deg Flush Shear Head
NAS 1767	Bolt, Close Tolerance, 100 Deg Reduced Head, Hi-Torque, Recess
NAS 1953-1970	Bolt, Shear, Hexagon Head, 180 KSI
NAS 1972-1980	Bolt, Flat, 100° Head Torq-Set and Hi-Torque, 180 KSI
NAS 1982-1990	Bolt, Brazier Head, Torq-Set and Hi-Torque, 180 KSI
NAS 1992-2000	Bolt, Flat, 100° Reduced Head, Torq-Set and Hi-Torque 180 KSI Shear
NAS 4002	Fasteners, Alloy Steel, Externally Threaded
NAS 4003	Fasteners, A286 CRES, Externally Threaded
NAS 4004	Fasteners, 6Al-4V Titanium Alloy, Externally Threaded
NAS 6203-6220	Bolt, Hex Head, Close Tolerance, Alloy Steel, Long Thread, Self-Locking and Non-Locking
NAS 6303-6320	Bolt, Hex Head, Close Tolerance, A286, Short Thread, Self-Locking and Non-Locking
NAS 6403-6420	Bolt, Hex Head, Close Tolerance, GAL-4V Titanium Alloy, Short Thread, Self-Locking and Non-Locking
NAS 6604-6620	Bolt, Hex Head, Close Tolerance, Alloy Steel, Long Thread, Self-Locking and Non-Locking
NAS 6704-6720	Bolt, Hex Head, Close Tolerance, A286 CRES, Long Thread, Self-Locking and Non-Locking
NAS 6804-6820	Bolt, Hex Head, Close Tolerance, GAL-4V Titanium Alloy, Long Thread, Self-Locking and Non-Locking
AMS 7478	Bolts and Screws, Steel, Corrosion and Heat Resistant Heat Treated, Roll Threaded-1800°F (982.2°C) Solution and Precipitation Heat Treated
AMS 7479	Bolts and Screws, Steel, Corrosion and Heat Resistant Heat Treated, Roll Threaded-1650°F (898.9°C) Solution and Precipitation Heat Treated

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3. Design and usage limitations

3.1 Bolts smaller than 0.2500 inch in diameter shall not be used in any single bolted structural connection, including primary flight control systems, or any application where failure would adversely affect safety of flight.

3.2 Aluminum alloy threaded bolts shall not be used in structural applications. (Does not apply to threaded pins. See Requirement 214.)

3.3 The smallest acceptable diameter for 100° reduced flush head bolts shall be 0.2500 inch, except 0.1900 inch dimater 100° reduced head bolts may be used in panels whose removal is not required for scheduled maintenance.

3.4 Silver plated or cadmium plated bolts shall not be used in contact with titanium structure.

3.5 Titanium alloy bolts shall not be used with silver plated self-locking nuts at temperatures above 600°F.

3.6 Titanium alloy bolts shall not be cadmium plated and shall not be used with cadmium plated nuts at temperatures above 200°F.

3.7 Cadmium plated bolts shall not be used in temperature probes, electrical or life support space vehicle components or systems, potable water supplies, or food processing equipment.

3.8 The self-locking element used in bolts shall be in accordance with MIL-F-8961 or MIL-F-18240. Self-locking bolts shall be used subject to the limitations stated herein.

a. Shall be selected and used in a manner that will permit functional and dimensional interchangeability with a part that has only the attributes described and defined by the applicable standard and drawings.

b. Shall be used only in applications that permit engagement with the complete internal threads over the minimum external thread as designated by X Minimum as shown on figure 201-1.

c. Bolts containing self-locking elements in accordance with MIL-F-18240 or an insert or part that is nonmetallic shall not be used in parts where the locking element will encounter keyways, slots, cross holes or other thread interruptions.

d. Shall not be used as stated herein:

(1) at joints in control systems, at single attachments or where the loss of the bolt would affect safety of flight

(2) as an axis of rotation for another part unless the fastener is held in a positive locking device that requires shearing or rupture of material before torsional loads would be applied to the bolt in such a manner as to relieve the internal stress of the assembly or turn the bolt loose.

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Examples: bearings, bushings, clamp-up bushings, pulleys, cranks, levers, linkages, hinge pins, axles, shafts, spindle gears, cams, cam followers, sliding mechanisms and pivot points

(3) at any single bolted structural joint which serves as a primary load path, the failure of which would endanger the safety of personnel or would render the equipment inoperative or cause its destruction.

Examples: fixed joints, tie rods, struts (fixed length members) wing attachments of fuselage, stabilizer surface attachments, longeron joints

e. Shall not be used to attach access panels, doors, or to assemble any parts that are routinely disassembled at intervals less than 400 flight hours.

f. Shall not be used on jet engine aircraft in locations where a loose fastener could fall or be drawn into the engine intake.

g. Shall not be used if the locking device has been reworked or reprocessed.

h. Self-locking bolts containing self-locking elements classified as 250°F, 450°F or 1200°F are intended for use at ambient temperature conditions of -65°F to +250°F, 450°F or 1200°F respectively and are designed to function satisfactorily at temperatures through these ranges.

i. When used in applications requiring controlled torque, such as clamping molded gaskets in fuel cells, consideration must be given to the maximum and minimum locking torque permitted by the procurement document.

j. For the self-locking bolts that incorporate an insert or part that is nonmetallic, the entering end of the threaded holes used in conjunction with the self-locking externally threaded fastener shall be countersunk 90 to 120 degrees. This countersink shall have a minimum diameter .015 inch larger than the major thread diameter of the fastener. This is to prevent the first thread from cutting the self-locking element.

k. Unthreaded holes or portions of holes through which the locking element of the bolt must pass shall have a minimum diameter sufficient to clear the locking element.

l. Shall not be used with castellated nuts or self-locking nuts.

4. Design selection and callout. Bolts in the category established by this requirement shall conform to the documents specified in section 2 and shall be selected from and specified by part numbers listed on the documents in tables 201-I and 201-II.

5. Tools and installation procedures

a. Interference between the hole and the head to shank radius should be avoided. A countersink in the bolt hole or the washer are methods commonly used.

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b. Bits for installing internal recess drive bolts shall be in accordance with MIL-B-9946, MS33750, MS33781 or MS14191, as applicable.

5.3 Wrenches for installing spline drive external wrenching bolts shall be in accordance with MIL-W-8982; hex head bolts shall be in accordance with MIL-W-8982 or GGG-W-1437.

5.4 Since the use of the improper wrench or bit will degrade installation, care shall be exercised to assure that specified size bit or wrench is available and used.

6. Intended use and guidance criteria

6.1 For the purposes of this requirement the terms bolt and screw are used interchangeably.

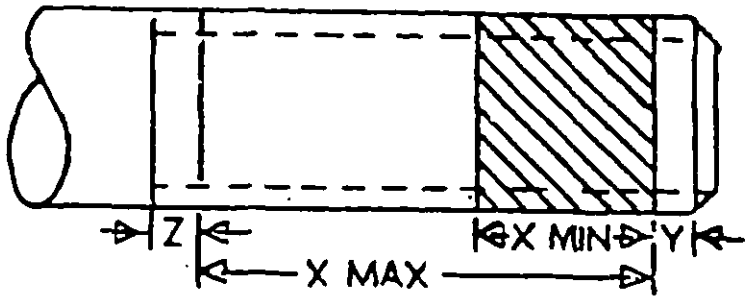
6.2 Where possible and practical, mating parts (except where flush head bolts or plate nuts are used) shall have similar external wrenching configurations.

6.3 When outgassing characteristics are a requirement, refer to Requirements 102 and 103 and to Marshall Space Flight Center Document 10M90001.

6.4 For the use of bolt threads in bearing refer to Requirement 121.

6.5 For the use of oversize bolts refer to Requirement 223.

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Thread Size	X Min.	Thread Size	X Min.	Thread Size	X Min.
4-40	.125	3/8-24	.208	7/8-14	.357
6-32	.156	7/16-20	.250	1-12	
8-32		1/2-20		1-1/8-12	.417
10-32		9/16-18	.278	1-1/4-12	
1/4-28	.178	5/8-18			
5/16-24	.208	3/4-16	.312		

X, min represents the minimum length of external threads required for engagement with complete internal thread pitch. The locking element shall engage within this minimum length and meet requirements of the locking element or fastener specification or standard approved for use. X, min is equal to 5 thread pitches.

X, max equals the length of complete thread pitches between Y and Z.

Y, is for ease in starting, the locking element shall not be effective within the area of Y minimum.

Y, min equals one complete thread pitch; Y, max equals two complete thread pitches. This dimension shall not include thread pitches which have incomplete form or unthreaded portions of end, that is, chamfer and any extension beyond thread.

Z, equals one complete thread pitch plus thread runout. The locking element or any machine holes or grooves for the locking element shall not penetrate this area.

FIGURE 201-1. Effective locking area.

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TABLE 201-I. Non-locking screws and bolts - continued

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC	FINISH 1/	OTHER INFO
			220 KSI FTU Tension			
Protruding	12 Spline	Steel	MS14163	MIL-B-8906	VAC Cadmium	Wheel Bolt
Protruding	12 Spline	Steel	MS21297	MIL-B-8906	VAC Cadmium	
Protruding	12 Spline	INCO 718	MS14181	MIL-B-8906	Bare	
			260 KSI FTU Tension			
Protruding	12 Spline	Steel	MS21296	MIL-B-8907	VAC Cadmium	
			95 KSI FSU Shear			
100° Flush Reduced	Hi-Torque	Steel	NAS1767	NAS498	Cadmium/ Aluminum	Crown Head
100° Flush Reduced	Hi-Torque	A286	NAS1767C	AMS7479	Passivated/ Cad/Alum	Crown Head2/
100° Flush Reduced	Hi-Torque	Ti	NAS1767V	NAS621	Bare/ Aluminum	Crown Head2/
100° Flush Reduced	Torq-Set	Steel	NAS1581A	NAS498	Cadmium	Use Only2/ Torq-Set Recess
100° Flush Reduced	Torq-Set	A286	NAS1581C	AMS7479	Passivated	Hi-Torque See NAS1767
100° Flush Reduced	Torq-Set	Ti	NAS1581V	NAS621	Bare	
100° Flush	Ribbed	Steel	MIL-B-87114/1	MIL-B-87114	Cadmium	Long Thread for use with MS14179
100° Flush Reduced	Torq-Set	Steel	MIL-B-87114/2	MIL-B-87114	Cadmium	
Pan		Steel	MIL-B-87114/3	MIL-B-87114	Cadmium	

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TABLE 201-I. Non-locking screws and bolts - continued

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC	FINISH 1/	OTHER INFO
108 KSI FSU						
100° Flush Reduced	Hi-Torque/ Torq-Set	Steel	NAS1992-2000	NAS4002	Cadmium	2/
100° Flush Reduced	Hi-Torque/ Torq-Set	A286	NAS1992C-2000C	NAS4003	Passivate/ Aluminum	2/
100°	Hi-Torque Torq-Set	Steel	NAS1972-1980	NAS4002	Cadmium	No Fatigue Requirement
100° Flush	Hi Torque/ Torq-Set	A286	NAS1972C-1980C	NAS4003	Passivated/ Aluminum	
132 KSI FSU						
Protruding	Spline	Steel	MIL-B-8906/2	MIL-B-8906	VAC Cadmium	

1/ Only finish acceptable.
2/ See 3.3 for limitation on use of #10 and smaller.

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TABLE 201-II. Self-locking bolts and screws.

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC 1/	FINISH 2/	TEMP °F	OTHER INFO
			160 KSI FTU				
Protruding	Hex	Steel	NAS6203L-6220L	NAS4002	Cadmium	250	Medium Thread
Protruding	Hex	A286	NAS6303L-6320L	NAS4003	Passivated	250	Medium Thread
Protruding	Hex	Ti	NAS6403L-6420L	NAS4004	Bare	250	Medium Thread
Protruding	Hex	Steel	NAS6603L-6620L	NAS4002	Cadmium	250	4/
Protruding	Hex	A286	NAS6703L-6720L	NAS4003	Passivated/ Cad/Alum	250	4/
Protruding	Hex	Ti	NAS6803L-6820L	NAS4004	Bare/ Aluminum	250	4/ 5/
			95 KSI FSU				
100° Flush	Hi-Torque	Steel	NAS1221L	NAS498	Cadmium	250	
100° Flush	Hi-Torque	A286	NAS1221EL	AMS7479	Passivated	250	
100° Flush	Hi-Torque	Ti	NAS1221VL	NAS621	Bare	250	
100° Flush	Torq-Set	Steel	NAS1161-1168	NAS498	Cadmium	250	
100° Flush	Torq-Set	A286	NAS1161E-1168E	AMS7478	Passivated	250	
100° Flush	Torq-Set	Ti	NAS1161V-1168V	NAS621	Bare	250	
100° Flush	Torq-Set	Steel	NAS1189	NAS498	Cadmium	250	Full Thread 3/
100° Flush	Torq-Set	A286	NAS1189E	AMS7478	Passivated	250	Full Thread 3/
100° Flush	Torq-Set	Ti	NAS1189V	NAS621	Bare	250	Full Thread 3/
100° Flush	Hi-Torque	Steel	NAS1219L	NAS498	Cadmium	250	Full Thread
100° Flush	Hi-Torque	A286	NAS1219EL	AMS7479	Passivated	250	Full Thread
100° Flush	Hi-Torque	Ti	NAS1219VL	NAS621	Bare	250	Full Thread

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TABLE 201-II. Self-locking bolts and screws. - continued

HEAD	DRIVE	MATERIAL	P/N	PROC SPEC 1/	FINISH 2/	TEMP °F	OTHER INFO
Pan	Hi-Torque	Steel	NAS1578AL	NAS496	Cadmium	250	
Pan	Torq-Set	A286	NAS1578CL	NAS496	Passivated	250	
Pan		Ti	NAS1578VL	NAS621	Bare	250	
Pan	Hi-Torque	Steel	NAS1218L	NAS498	Cadmium	250	Sizes .1120-40
Pan	Hi-Torque	A286	NAS1218EL	AMS7479	Passivated	250	Thru .1640-32;
Pan	Hi-Torque	Ti	NAS1218VL	NAS621	Bare	250	for .1900 and larger see NAS1578
Pan	Torq-Set	Steel	NAS1190	NAS498	Cadmium	250	Full Thread 3/
Pan	Torq-Set	A286	NAS1190E	AMS7478	Passivated	250	Full Thread 3/
Pan	Torq-Set	Ti	NAS1190V	NAS621	Bare	250	Full Thread 3/
Pan	Hi-Torque	Steel	NAS1216L	NAS498	Cadmium	250	Full Thread
Pan	Hi-Torque	A286	NAS1216EL	AMS7479	Passivated	250	Full Thread
Pan	Hi-Torque	Ti	NAS1216VL	NAS621	Bare	250	Full Thread

1/ Non-metallic locking feature in accordance with MIL-F-18240 or as specified on part standard.
2/ Only finish acceptable.
3/ Torq-Set recess only approved for use.
4/ See standard for exception to procurement specification.
5/ Cadmium plated bolt not approved.

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STUDS

1. Purpose. This requirement provides the engineering criteria, requirements and limitations for the selection and use of studs as aerospace mechanical fastening systems.

2. Documents applicable to Requirement 202

2.1 Government documents

Military Specifications

MIL-S-45933	Stud, Keyring Locked, 125 and 160 KSI, FTU: General Specifications for
MIL-S-45933/1	Stud, Keyring Locked, 125 KSI, FTU, Single Step
MIL-S-45933/2	Stud, Keyring Locked, 160 KSI, FTU, Multiple Step
MIL-S-45933/3	Stud, Keyring Locked, Hole Dimensions for and Assembly of

Military Standards

MS 9827	Stud - Stepped, 2 DIA, Engagement, Steel, .250-20 UNJS x .190-32 UNJF
MS 9828	Stud - Stepped, 2 DIA, Engagement, Steel, .3125-18 UNJS x .250-28 UNJF
MS 9829	Stud - Stepped, 2 DIA, Engagement, Steel, .375-16 UNJS x .3125-24 UNJF
MS 9830	Stud - Stepped, 2 DIA, Engagement, Steel, .4375-14 UNJS x .375-24 UNJF
MS 9831	Stud - Stepped, 2 DIA, Engagement, Steel, .500-13 UNJS x .4375-20 UNJF
MS 9832	Stud - Stepped, 2 DIA, Engagement, Steel, .5625-12 UNJS x .500-20 UNJF
MS 9833	Stud - Stepped, 2 DIA, Engagement, Steel, .625-11 UNJS x .5625-18 UNJF
MS 9834	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .250-20 UNJS x .190-32 UNJF
MS 9835	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .3125-18 UNJS x .250-28 UNJF
MS 9836	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .375-16 UNJS x .3125-24 UNJF
MS 9837	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .4375-14 UNJS x .375-24 UNJF
MS 9838	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .500-13 UNJS x .4375-20 UNJF
MS 9839	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .5625-12 UNJS x .500-20 UNJF
MS 9840	Stud - Stepped, Drilled, 2 DIA, Engagement, Steel, .625-11 UNJS x .5625-18 UNJF
MS 17293	Stud - Stepped, 1.5 DIA, Engagement, .250-20 x .190-32 (MIL-S-8879 Thread)
MS 17294	Stud - Stepped, 1.5 DIA, Engagement, .3125-18 x .250-28 (MIL-S-8879 Thread)

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MS 17295	Stud - Stepped, 1.5 DIA, Engagement, .375-16 x .3125-24 (MIL-S-8879 Thread)
MS 17296	Stud - Stepped, 1.5 DIA, Engagement, .4375-14 x .375-24 (MIL-S-8879 Thread)
MS 17297	Stud - Stepped, Drilled, 1.5 DIA, Engagement, .3125-18 x .250-28 (MIL-S-8879 Thread)
MS 17298	Stud - Stepped, Drilled, 1.5 DIA, Engagement, .375-16 x .3125-24 (MIL-S-8879 Thread)
MS 17299	Stud - Straight, .190-24 x .190-32 (MIL-S-8879 Thread)
MS 17300	Stud - Straight, .250-20 x .250-28 (MIL-S-8879 Thread)
MS 17301	Stud - Straight, .3125-18 x .3125-24 (MIL-S-8879 Thread)
MS 17302	Stud - Straight, Drilled, .250-20 x .250-28 (MIL-S-8879 Thread)
MS 17303	Stud - Straight, Drilled, .3125-18 x .3125-24 (MIL-S-8879 Thread)
MS 51833	Stud, Locked In-Key Locked, Lightweight
MS 51834	Stud, Locked In-Key Locked, Heavy Duty
MS 51835	Inserts and Studs, Locked In-Key Locked, Hole Dimensions for and Assembly of
MS 51989	Stud, Locked In-Ring Locked, Serrated
MS 51990	Ring, Lock, Serrated
MS 51992	Stud, Locked In-Ring Locked, Serrated, High Strength
MS 51995	Fasteners, Ring Locked Inserts and Studs, Installation of
MS 51997	Ring, Lock, Serrated - High Strength

Military Handbooks

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures

2.2 Non-Government documents

NAS1841 thru NAS1854	Stud Assembly - Threaded Metal, Double Step
NAS1860 thru NAS1863	Stud Assembly - Threaded Metal, Single Step
AS1229	Insert, Screw Thread, Helical Coil, Stud Locking, Performance Standard for
AS3086	Stud - Straight, Key Locked, Steel, .190-32 UNJF x .190-32 UNJF
AS3087	Stud - Straight, Key Locked, Steel, .250-28 UNJF x .250-28 UNJF
AS3088	Stud - Straight, Key Locked, Steel, .3125-24 UNJF x .3125-24 UNJF
AS3089	Stud - Straight, Key Locked, Steel, .375-24 UNJF x .375-24 UNJF
AS3090	Stud - Straight, Key Locked, CRES AMS5731, .190-32 UNJF x .190-32 UNJF
AS3091	Stud - Straight, Key Locked, CRES AMS5731, .250-28 UNJF x .250-28 UNJF
AS3092	Stud - Straight, Key Locked, CRES AMS5731, .3125-24 UNJF x .3125-24 UNJF
AS3093	Stud - Straight, Key Locked, CRES AMS5731, .375-24 UNJF x .375-24 UNJF
AS3139	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .250-28 UNJF - 3A x .190-32 UNJF - 3A
AS3140	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .3125-24 UNJF - 3A x .250-28 UNJF - 3A

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AS3141	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .375-24 UNJF - 3A x .3125-24 UNJF - 3A
AS3142	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .4375-20 UNJF - 3A x .375-24 UNJF - 3A
AS3143	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .500-20 UNJF - 3A x .4375-20 UNFJ - 3A
AS3144	Stud, Stepped - 2.5 DIA. Engagement, PD Shank, AMS5731, .5625-18 UNJF - 3A x .580-20 UNJF - 3A
AS3319	Stud - Straight, Ring Locked, CRES AMS5731, .190-24 UNJC x .190-32 UNJF
AS3320	Stud - Straight, Ring Locked, CRES AMS5731, .250-20 UNJC x .190-32 UNJF
AS3321	Stud - Straight, Ring Locked, CRES AMS5731, .3125-18 UNJC x .3125-24 UNJF
AS3322	Stud - Straight, Ring Locked, CRES AMS5731, .375-16 UNJC x .375-24 UNJF

3. Design and usage limitations. Studs for tapped holes shall be subject to the limitations and usages stated herein:

a. Studs shall be installed as specified in their particular design/usage standard to eliminate the possibility of their rotational displacement when installing or removing the mating unit.

b. Install MS 51833 and MS 51834 studs in accordance with MS 51835.

c. Install MIL-S-45933/1 and MIL-S-45933/2 studs in accordance with MIL-S-45933/3.

d. Install MS 51989 and MS 51992 studs in accordance with MS 51995.

e. Install MS 9827 thru MS9840 and MS 17293 thru MS 17303 studs in threaded holes tapped to National Form NC-3 tolerances or Unified National Form UNC-3B tolerances to the required depth. The studs shall be installed to the torque limit specified in table 202-I (for MS 17299 thru MS 17303) and table 202-II (for MS 9827 thru MS 9840 and MS 17293 thru MS 17298). In order to meet the required installation torque limits given in tables 202-I and 202-II, a standard or a +.003 oversize pitch diameter stud may be selected.

f. Install AS3086 thru AS3093 studs in accordance with the manufacturer's instrutions. Refer to standards AS3086 thru AS3093 for manufacturer's name and address.

g. Install AS3139 thru AS3144 studs into stud-locking helical coil threaded inserts (per AS1229) designed to proved an installation torque within the values specified in table 202-III.

h. Nut end threads of studs under.250 nominal diameter shall not be drilled for cotter pins or lock wire.

4. Design, selection and approved call-out. Fasteners in the category established by this requirement shall be selected and called-out from table 202-IV.

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5. Tooling. Studs shall be installed by using tools that conform to acceptable aerospace practices for the type of stud being installed.

6. Intended use and guidance criteria. This section is intended to provide guidance to designers for the application and selection of studs.

a. Stepped studs may be used to provide higher strength capablities in relatively low strength structural (parent) materials with a shorter length of engagement than required for straight studs (both ends same diameter).

b. In the design selection of studs for structural applications, it is necessary that the proper degree of consideration be given to the axial load carrying capabilities of the installed stud in the specific parent materials at the maximum operating temperature of the assembly.

c. The axial load capability of an installed stud is determined by either the minimum tensile strength of the nut end or by the resistance to pullout from the parent material whichever is the lower.

(1) Resistance to pullout of the installed stud is the product of the shear engagement area of the stud end thread and the allowable shear stress of the parent material at the maximum operating temperature. It does not represent a dimension of either of the members in an unassembled condition.

(2) The allowable shear stress (Fs) for most metallic materials is listed in MIL-HDBK-5.

(3) The shear engagement areas of the stud end threads may be obtained from the design/usage standard for the particular stud.

d. Most studs are intended to be loaded primarily in tension. Joints carrying shear loads should be designed to preclude subjecting the studs to shear loads or use only those studs whose features allow their use under shear loading.

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TABLE 202-I. Straight interference fit studs.

Thread size (Drive end)	Torque, lb - in.	
	Minimum	Maximum
.250 -20	40	105
.3125 -18	85	230

TABLE 202-II. Stepped interference fit studs.

Thread size (nut end)	Torque, lb - in.	
	Minimum	Maximum
.190 -32	15	50
.250 -28	40	125
.3125 -24	85	260
.375 -24	160	500
.4375 -20	200	800
.500 -20	250	1300
.5625 -18	425	1800

TABLE 202-III. Stepped studs (AS3139 to AS3144) in stud-locking helical coil inserts per AS1229.

Thread size (nut end)	Torque, lb - in. 1/	
	Minimum	Maximum
.190 -32	23	45
.250 -28	52	90
.3125 -24	105	180
.375 -24	140	240
.4375 -20	175	300
.500 -20	260	450

1/ A lubricant or coating may be applied to the stud end threads so as not to exceed the maximum torque values.

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TABLE 202-IV. Studs.

Part Number	Procurement Specification	Material	Other
Stud - Stepped 1.5 Dia Engagement			
MS 17293	None	Steel	.250-20 x .190-32
MS 17294	None	Steel	.3125-18 x .20-28
MS 17295	None	Steel	.375-16 x .3125-24
MS 17296	None	Steel	.4375-14 x .375-24
MS 17297	None	Steel	.3125-18 x .250-28
MS 17298	None	Steel	.375-16 x .3125-24
Stud - Stepped 2.0 Dia Engagement			
MS 9827	AMS 7456	Steel	.250-20 x .190-32
MS 9828	AMS 7456	Steel	.3125-18 x 250-28
MS 9829	AMS 7456	Steel	.375-16 x .3125-24
MS 9830	AMS 7456	Steel	.4375-14 x .375-24
MS 9831	AMS 7456	Steel	.500-13 x .4375-20
MS 9832	AMS 7456	Steel	.5625-12 x .50-20
MS 9833	AMS 7456	Steel	.625-11 x .5625-18
MS 9834	AMS 7456	Steel	.250-20 x .190-32
MS 9835	AMS 7456	Steel	.3125-18 x .250-28
MS 9836	AMS 7456	Steel	.375-16 x .3125-24
MS 9837	AMS 7456	Steel	.4375-14 x .375-24
MS 9838	AMS 7456	Steel	.500-13 x .4375-20
MS 9839	AMS 7456	Steel	.5625-12 x .500-20
MS 9840	AMS 7456	Steel	.625-11 x .5625-18

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TABLE 202-IV. Studs. - Continued

Part Number	Procurement Specification	Material	Other
Stud - Stepped 2.5 Dia Engagement			
AS 3139	AMS 7477	CRES (AMS 5731)	.250-28 x .190-32
AS 3140	AMS 7477	CRES (AMS 5731)	.3125-24 x .250-28
AS 3141	AMS 7477	CRES (AMS 5731)	.375-24 x .3125-24
AS 3142	AMS 7477	CRES (AMS 5731)	.4375-20 x .375-24
AS 3143	AMS 7477	CRES (AMS 5731)	.500-20 x .4375-20
AS 3144	AMS 7477	CRES (AMS 5731)	.5625-18 x .580-20
Stud - Straight			
MS 17299	None	Steel	.190-24 x .190-32
MS 17300	None	Steel	.250-20 x .250-28
MS 17301	None	Steel	.3125-18 x .3125-24
MS 17302	None	Steel	.250-20 x .250-28
MS 17303	None	Steel	.3125-18 x .3125-24
AS 3086	AMS 7452	Steel	.190-32 x .190-32
AS 3087	AMS 7452	Steel	.250-28 x .250-28
AS 3088	AMS 7452	Steel	.3125-24 x .3125-24
AS 3089	AMS 7452	Steel	.375-24 x .375-24
AS 3090	AMS 7477	CRES (AMS 5731)	.190-32 x .190-32
AS 3091	AMS 7477	CRES (AMS 5731)	.250-28 x .250-28
AS 3092	AMS 7477	CRES (AMS 5731)	.3125-24 x .3125-24
AS 3093	AMS 7477	CRES (AMS 5731)	.375-24 x .375-24
AS 3319	AMS 7482	CRES (AMS 5731)	.190-24 x .190-32
AS 3320	AMS 7482	CRES (AMS 5731)	.250-20 x .250-28
AS 3321	AMS 7482	CRES (AMS 5731)	.3125-18 x .3125-24
AS 3322	AMS 7482	CRES (AMS 5731)	.375-16 x .375-24

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TABLE 202-IV. Studs. - Continued

Part Number	Procurement Specification	Material	Other
Stud - Keyring Locked			
M 45933/1	MIL-S-45933	Steel (4130, 4140, A 286)	125 ksi FTU
M 45933/2	MIL-S-45933	Steel (4130, 4140, A 286)	160 ksi FTU
MS 51833	MIL-S-45915	Steel (4140, A 286, 303)	Lightweight
MS 51834	MIL-S-45915	Steel (4140, A 286, 303)	Heavy Duty
MS 51989	MIL-S-45909	Steel (4130, A 286)	
MS 51992	MIL-S-45909	Steel (8740, A 286) Nickel (AMS 5662D-79) Titanium (6Al-4V)	High Strength
Ring, Locked, Serrated			
MS 51990	MIL-I-45910	Steel (C1117, A 286)	
MS 51997	MIL-I-45910	Steel (C1117, A 286)	
Stud Assembly			
NAS 1841 thru NAS 1854	None	Steel (4130, 303, A 286)	Double Step
NAS 1860 thru NAS 1863	None	Steel (4130, 303, A 286)	Single Step

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SOLID RIVETS

1. Purpose. This requirement provides the engineering criteria, requirements, and design and usage limitations for the selection and use of solid rivets in aerospace mechanical fastening systems.

2. Documents applicable to Requirement 206

2.1 Government documents

Military Standards

MS 9403	Rivet, Solid-Universal Head, AMS5737
MS 9460	Rivet, Solid - 100° Flush Head, AMS7235
* MS 14218	Rivet, Solid, 120° Flush Interference Shear Head
MS 20426	Rivet, Solid, Countersunk 100°, Precision Head, Aluminum and Aluminum Alloy
MS 20427	Rivet, Solid 100° Countersunk Head, Carbon Steel Corrosion-Resistant Steel, Monel and Copper
MS 20470	Rivet, Solid-Universal Head, Aluminum and Aluminum Alloy
MS 20613	Rivet, Solid-Universal Head, Steel, Carbon and Steel, Corrosion-Resistant
MS 20615	Rivet, Solid-Universal Head, Brass, Copper and Nickel-Copper Alloy

Air Force - Navy Aeronautical Standards

AN123151 thru AN123300	Rivet, Solid, Universal Head, AMS7229
AN123301 thru AN123450	Rivet, Solid, Universal Head, AMS7232
AN123451 thru AN123600	Rivet, Solid, 100° Flush Head, AMS7229
AN123601 thru AN123750	Rivet, Solid, 100° Flush Head, AMS7232

2.2 Non-Government documents

* NAS1097	Rivet, Solid, 100° Flush Shear Head, Aluminum Alloy, Titanium Columbium Alloy
NAS1198	Rivet - Solid Universal Head, A286 Corrosion Resistant Steel
NAS1199	Rivet - Solid - 100° Flush Head A286 Corrosion Resistant Steel
NAS1200	Rivet, Solid, 100° Flush Shear Head A286 Corrosion Resistant Steel and Monel

3. Design and usage limitations. Solid rivets in this requirement shall be * subject to the design and usage limitations stated herein.

a. Do not use cadmium-plated rivets where temperatures may exceed 450°F. Cadmium-plated monel rivets may not be used in applications where temperatures exceed 400°F.

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b. Do not use cadmium-plated or silver-plated rivets in titanium.

* c. The primary loading of rivets shall be shear.

d. Head configuration shall be in accordance with Requirement 106. Where an angle other than 100° is needed, contact the procuring activity for approval of the specific application.

* e. Edge distance (center of hole to edge of sheet) for the location of rivets in sheets shall be a minimum of 2D (D = rivet diameter).

4. Design, selection and approved call-out. Fasteners in the category established by this requirement shall be selected from and called out by part numbers specified on standards listed in section 2.

5. Tooling. Rivets shall be driven utilizing tools that conform to acceptable aerospace practices for the rivet size and material being upset.

* 6. Intended use and guidance criteria. This section is intended to provide guidance to designers for the application and selection of solid rivets.

6.1 Aluminum alloy 2024 ("Type DD") rivets should be driven immediately after quenching or refrigeration. These require special care when driving to avoid or minimize upset cracking. See figure 206-1 for allowances.

6.2 When using rivet material harder than the material to be joined, particular care should be taken to avoid distortion during riveting. Special care is recommended when selecting rivet types and materials for installation through nonmetallic structures. Soft materials may be riveted by using washers under the rivet upset trail.

6.3 To minimize galvanic corrosion in the joint, rivets should not be anodic to the most anodic material in the joint.

6.4 Spotfacing should be used to provide a flat surface under upset heads when:

6.4.1 The surface slope is greater than 8° under the upset head of rivets. (Because of fatigue considerations, tapered shims may be used to avoid counterbores.)

6.4.2 A curved surface has a radius less than three times the rivet shank diameter.

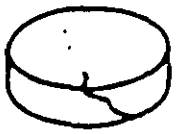
6.4.3 The roughness of the facing surface under the heads is greater than 500 RHR.

6.5 Do not use solid rivets where forces required to upset the solid rivet can be detrimental to the structure.

6.6 Minimum spacing for riveted joints should be 4D between fasteners in the same adjacent rows or in staggered patterns except in fuel tight areas where the minimum spacing should be 3D (D = rivet diameter).

6.7 Tables 206-I and 206-II are presented as an aid in the selection of solid rivets.

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Acceptable provided cracks do not extend within a circle concentric with and having a diameter approximately 1.1 times the shank diameter.



Acceptable provided cracks do not extend within a circle concentric with and having a diameter approximately 1.1 times the shank diameter and provided the cracks do not tend to intersect so as to be a potential cause of a section of the head chipping out.

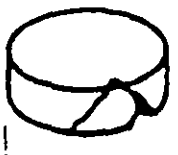


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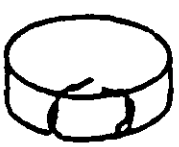
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REQUIREMENT 206
24 June 1983



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Not Acceptable

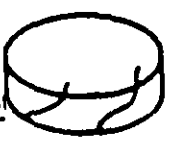


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FIGURE 206-1. Formed head inspectio

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TABLE 206-I. Rivet, solid, flush head.

PART NUMBER	MATERIAL	FINISH	ROOM TEMPERATURE ULTIMATE SHEAR STRENGTH, AS RECEIVED, PSI	MAXIMUM * TEMPERATURE
100° COUNTERSUNK				
AN123451 thru AN123600	CRES, 300 Series	None	48,000 (MIN)	600°F
AN123601 thru AN123750	Ni Alloy	None	51,000 (MIN)	1800°F
MS9460-( )	CRES, A-286	None	80,000 to 95,000	1200°F
*MS14218	Al Alloy	Anodized or Chemically Surface Treated	24,000 to 38,000	250°F
MS20426A( )-( )	Al, 1100-F	None	No Shear Test Required	250°F
MS20426B( )-( )	Al, 5056-1132	Anodized or Chemically Surface Treated	24,000 (MIN)	250°F
MS20426AD( )-( )	Al, 2117-T4	Anodized or Chemically Surface Treated	26,000 (MIN)	250°F
MS20426DD( )-( )	Al, 2024-T4	Anodized	37,000 (MIN)	250°F
MS20427-( )C( )	Steel, C1010-C1015	Cd Plated	32,000 to 38,000	450°F

* Denotes change.

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TABLE 206-I. Rivet, solid, flush head. - Continued

PART NUMBER	MATERIAL	FINISH	ROOM TEMPERATURE ULTIMATE SHEAR STRENGTH, AS RECEIVED, PSI	MAXIMUM * TEMPERATURE
MS20427F( )-( )	CRES, 300 Series	None	65,000 to 85,000	600°F
MS20427M( )-( )	Monel	None	49,000 to 59,000	800°F
MS20427M( )C( )	Monel	Cd Plated	49,000 to 59,000	400°F
MS20427C( )-( )	Copper, Annealed	None	23,000 to 27,000	300°F
NAS1199-( )-( )	CRES, A-286	Passivated	85,000 to 95,000	1200°F
100° SHEAR HEAD NAS1097B( )-( )	Al, 5056-1132	Anodized or Chemically Surface Treated	24,000 (MIN)	250°F
NAS1097AD( )-( )	Al, 2117-T4	Anodized or Chemically Surface Treated	26,000 (MIN)	250°F
NAS1097D( )-( )	Al, 2017-T4	Anodized or Chemically Surface Treated	33,000 (MIN)	250°F
NAS1097DD( )-( )	Al, 2024-T4	Anodized	37,000 (MIN)	250°F
AS1200-( )-( )	CRES, A-286	Passivated	85,000 to 95,000	1200°F
NAS1200M( )-( )	Monel	None	49,000 to 59,000	800°F
NAS1200M( )-( )P	Monel	Cd Plated	49,000 to 59,000	400°F

Maximum temperature column is for information/guidance only. Some rivets may sustain strength for only a few seconds at these temperatures. Consequently, it is recommended that consideration be given to the time-at-temperature characteristics of each material specified in table 206-I to avoid severe degrading of rivets at maximum temperature exposure.

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TABLE 206-II. Rivet, solid, universal head.

PART NUMBER	MATERIAL	FINISH	ROOM TEMPERATURE ULTIMATE SHEAR STRENGTH, AS RECEIVED, PSI	MAXIMUM * TEMPERATURE
AN123151 thru AN123300	CRES, 300 Series	None	48,000 (MIN)	600°F
AN123301 thru AN123450	Ni Alloy	None	51,000 (MIN)	1800°F
MS9403-( )	CRES, A-286	None	80,000 to 95,000	1200°F
MS20470A( )-( )	Al, 1100-F	None	No Shear Test Required	250°F
MS20470B( )-( )	Al, 5056-1132	Anodized or Chemically Surface Treated	24,000 (MIN)	250°F
MS20470AD( )-( )	Al, 2117-T4	Anodized or Chemically Surface Treated	33,000 (MIN)	250°F
MS20470D( )-( )	Al, 2017-T4	Anodized or Chemically Surface Treated	33,000 (MIN)	250°F
*MS20470DD( )-( )	Al, 2024-T4	Anodized	37,000 (MIN)	250°F
MS20613-( )P( )	Steel, C1005-C1015	Cd Plated	25,000 to 42,000	450°F
MS20613-( )C( )	CRES, 300 Series	Passivated	65,000 to 85,000	600°F
MS20615-( )B( )	Brass, 1/8 Hard	None	35,000	300°F
MS20615-( )Cu( )	Copper, Annealed	None	17,000 to 30,000	300°F

* Denotes change.

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TABLE 206-II. Rivet, solid, universal head. - continued

PART NUMBER	MATERIAL	FINISH	ROOM TEMPERATURE ULTIMATE SHEAR STRENGTH, AS RECEIVED, PSI	MAXIMUM * TEMPERATURE
MS20615-( )M( )	Monel	None	49,000 to 59,000	800°F
MS20615-( )MP( )	Monel	Cd Plated	49,000 to 59,000	400°F
NAS1198-( )-( )	CRES, A-286	Passivated	85,000 to 95,000	1200°F

Maximum temperature column is for information/guidance only. Some rivets may sustain strength for only a few seconds at these temperatures. Consequently, it is recommended that consideration be given to the time-at-temperature characteristics of each material specified in table 206-II to avoid severe degrading of rivets at maximum temperature exposure.

* Denotes change.

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BLIND FASTENERS

1. Purpose. This requirement provides the engineering criteria, requirements, and design and usage limitations for the selection and use of blind fasteners in aerospace structure.

2. Documents applicable to Requirement 213

2.1 Government documents

Military Specifications

MIL-R-007885(AS)	Rivets; Blind, Structural, Locked and Friction Retained Spindle, General Specification For
MIL-R-007885/2	Rivets, Blind, Structural, Universal Protruding Head, 5056 Al Alloy Sleeve, Positive Locked, 8740 Alloy Steel Spindle (Type I, Style A, Class 1)
MIL-R-007885/3	Rivets, Blind, Structural, 100° Flush Head, 5056 Al Alloy Sleeve, Positive Locked, 8740 Alloy Steel Spindle (Type I, Style A, Class 2)
MIL-R-007885/4	Rivets, Blind, Structural, Universal Protruding Head, Monel Sleeve, Positive Locked 15-7PH CRES Spindle (Type I, Style A, Class 1)
MIL-R-007885/5	Rivets, Blind, Structural, 100° Flush Head, Monel Sleeve, Positive Locked 15-7PH CRES Spindle (Type I, Style A, Class 2)
MIL-R-007885/6	Rivets, Blind, Structural, Universal Protruding Head, 5056 Al Alloy Sleeve, Positive Locked, 8740 Alloy Steel Spindle, Oversize Diameter (Type I, Style B, Class 1)
MIL-R-007885/7	Rivets, Blind, Structural, 100° Flush Head, 5056 Al Alloy Sleeve, Positive Locked, 8740 Alloy Steel Spindle, Oversize Diameter (Type I, Style B, Class 2)
MIL-R-007885/8	Rivets, Blind, Structural, Universal Protruding Head, Monel Sleeve, Positive Locked, 15-7PH CRES Spindle, Oversize Diameter (Type I, Style B, Class 1)
MIL-R-007885/9	Rivets, Blind, Structural, 100° Flush Head, Monel Sleeve, Positive Locked, 15-7 PH CRES Spindle, Oversize Diameter (Type I, Style B, Class 2)
MIL-F-8975	Fasteners, Blind, High Strength, Installation Formed, Corrosion Resistant Steel, Heat Resistant Steel and Titanium, General Specification For
MIL-F-8975/2	Fasteners, Blind, High Strength, Corrosion Resistant Steel, Flush Head, Type II, Class 1
MIL-F-8975/3	Fasteners, Blind, Titanium, Flush Head, Type II, Class 2
MIL-F-8975/4	Fasteners, Blind, High Strength, Corrosion Resistant and Heat Resistant Steel, Protruding Head, Type II, Class 1
MIL-F-8975/5	Fasteners, Blind, Titanium, Protruding Head, Type II, Class 2
MIL-F-81177	Fasteners, Blind, High Strength, Installation Formed, Alloy Steel, General Specification For
MIL-F-81177/2	Fasteners, Blind, High Strength, Installation Formed, Threaded, Self-Locking, 100° Flush Head, 450°F, Alloy Steel, 112 KSI Fsu

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MIL-F-81177/3	Fasteners, Blind, High Strength, Installation Formed, Threaded, Self-Locking, Protruding Head, 450°F, Alloy Steel, 112 KSI Fsu
MIL-R-85188(AS)	Riveter, Power, Pneumatic-Hydraulic or Pneumatic Blind Rivet Installation

Military Standards

MS 20426	Rivet, Solid, Countersink 100°, Precision Head, Aluminum and Aluminum Alloy
MS 20470	Rivet, Solid-Universal Head, Aluminum and Aluminum Alloy
MS 21140	Fastener, Blind, High Strength, Pull Type, Positive Mechanical Lock, 100° Flush Head, Corrosion Resisting Steel, 95 KSI Fsu
MS 21141	Fastener, Blind, High Strength, Pull Type, Positive Mechanical Lock Protruding Head, Corrosion Resisting Steel, 95 KSI Fsu
MS 24694	Screw, Machine, Flat Countersunk Head, 100°, Structural, Cross Recessed, UNC-3A and UNF-3A
MS 90353	Fastener, Blind, High Strength, Pull Type, Positive Mechanical Lock, 100° Flush Head, Alloy Steel, 112 KSI Fsu
MS 90354	Fastener, Blind, High Strength, Pull Type, Positive Mechanical Lock, Protruding Head, Alloy Steel, 112 KSI Fsu

2.2 Non-Government documents

NAS 1097	Rivet, Solid, 100° Flush Shear Head, Aluminum Alloy, Titanium Columbium Alloy
NAS 1398	Rivet-Blind, Protruding Head, Locked Spindle
NAS 1399	Rivet-Blind, 100° Flush Head, Locked Spindle
NAS 1400	Rivet-Blind, Self-Plugging, Mechanically Locked Spindle
NAS 1669	Fastener-Blind, Internally Threaded, External Sleeve, General Purpose, Protruding Head, Self-Locking
NAS 1670	Fastener-Blind, Internally Threaded, External Sleeve, General Purpose, Flush Head, Slef-Locking
NAS 1671	Fastener-Blind, Internally Threaded, External Sleeve, High Temperature, Protruding Head, Self-Locking
NAS 1672	Fastener-Blind, Internally Threaded, External Sleeve, High Temperature, Flush Head, Self-Locking
NAS 1673	Fastener-Blind, Internally Threaded, External Sleeve, Lightweight, Protruding Head, Slef-Locking
NAS 1674	Fastener-Blind, Internally Threaded, External Sleeve, Lightweight, Millable Head, Self-Locking
NAS 1675	Fastener-Blind, Internally Threaded, External Sleeve, Self-Locking
NAS 1738	Rivet-Blind, Protruding Head, Mechanically Locked Spindle, Bulbed
NAS 1739	Rivet-Blind, 100° Flush, Mechanically Locked, Spindle, Bulbed
NAS 1740	Rivet-Blind, Self-Plugging, Mechanically Locked Spindle, Bulbed
NAS 1768	Rivet-Blind, Protruding Head, Mechanically Locked Spindle, Bulbed
NAS 1769	Rivet-Blind, 100° Flush Head, Mechanically Locked Spindle, Bulbed

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NAS 1900	Rivet, Blind-General Purpose, Bulbed, Self-Plugging, Mechanically-Locked-Spindle
NAS 1919	Rivet, Blind-General Purpose, Bulbed, Protruding Head, Mechanically-Locked-Spindle
NAS 1921	Rivet, Blind-General Purpose, Bulbed, 100° Flush Head, Mechanically-Locked-Spindle

3. Design and usage limitations

3.1 Blind fastener general design and usage limitation are as stated herein.

3.1.2 Blind fasteners are primarily intended for applications where inaccessibility precludes the use of conventional fasteners.

3.1.3 Cadmium plated blind fasteners shall not be used where the temperature is above 450°F.

3.1.4 Cadmium plated blind fasteners shall not be used in space vehicle components or systems due to the instability of cadmium in a vacuum environment. Blind fasteners that are lubricated with dry film lubricants may be used in space applications provided the lubricant has been approved as meeting the outgasing requirements.

3.1.5 Cadmium plated blind fasteners may be used in titanium up to 200°F. CAUTION - Embrittlement of titanium from cadmium is directly related to temperature and structural bearing stresses normally encountered in end grain situations.

3.1.6 Blind fasteners shall be used in structural joints only where shear loads are the primary dsign load consideration.

3.1.7 Blind fasteners shall not be used in liquid tight areas without acquisition activity engineering approval.

3.1.8 Flush head styles may be used in conjunction with either machine countersinking or machine sub-countersinking where the sheet next to the head is dimpled. Flush head styles shall not be used where the blind side sheet is dimpled.

3.1.9 Blind fasteners shall not be used in applications where they are subject to removal during routine servicing and overhaul.

3.1.10 Blind fasteners head configuration shall conform to requirements in existing documents which specify appropriate parameters.

3.1.11 All blind fasteners shall be capable of obtaining functional characteristics when installed in hole sizes specified in their appropriate design and usage limitation standards and procurement specifications.

3.1.12 The grip increments for blind fasteners shall conform to requirements in existing documents which specify appropriate parameters.

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3.1.13 Blind fasteners shall conform to shear strengths specified on their applicable specification or standard.

3.1.14 Unprotected areas of the blind fastener manufactured head shall receive a protective coating as necessary to comply with the aerospace vehicle finish requirements.

3.2 Blind rivet general design and usage limitations are as follows:

3.2.1 Blind rivets shall not be used on control surface hinges, hinge brackets, flight control actuating systems attachment, wing attach fittings, landing gear fittings or similar applications.

3.2.2 Mechanically locked spindle blind rivets may be used in engine inlet areas. Friction locked spindle blind rivets (no locking ring or collar) shall not be used in engine inlet areas.

3.2.3 Blind rivet holes for dimpled assembly shall be drilled to size after dimpling.

3.3 Blind bolt design and usage limitations are as stated herein.

3.3.1 Mechanically locked spindle blind bolts may be used in engine inlet areas. Friction locked spindle blind bolts (no locking ring or collar) shall not be used in engine inlet areas.

4. Design, selection and approved call out. Fasteners in the category established by this requirement shall be selected from and called-out by part numbers specified on standards or specifications slash sheets listed in table 213-I.

5. Tooling. Blind fasteners shall be installed by utilizing tools that are recommended by the fastener manufacturer. It is recommended that newly developed blind rivets be installed with MIL-R-85188 type tooling.

6. Intended use and guidance criteria. This section is intended to provide guidance for the application and selection of blind fasteners.

6.1 The head configuration of newly developed blind fasteners shall be protruding head or countersunk head. The countersunk head configuration shall be of the 100° included angle design in accordance with dimensions shown on MS 20426, MS 24694, or NAS 1097. The protruding head configuration shall be in accordance with dimensions shown on MS 20470, MS 90354 or NAS 1669.

6.2 All newly developed blind fasteners shall be capable of obtaining functional characteristics as required by applicable specifications or standards when installed in hole sizes for respective diameter dash numbers and types as listed in table 213-II.

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TABLE 213-I. Blind fasteners.

Part Number	Procurement Specification	Material	Other
Protruding Head			
MS 21141	MIL-F-8975	A 286	95 ksi FSU
MS 90354	MIL-F-81177	Alloy Steel	112 ksi FSU
NAS 1398	NAS 1400	Al (5056, 2017) Monel, A 286	
NAS 1669	NAS 1675	Alloy Steel (Nut & Screw) CRES (Sleeve)	
NAS 1671	NAS 1675	A 286	1200°F (max)
NAS 1673	NAS 1675	Al (7075) (Nut) Alloy Steel (Screw) CRES (Sleeve)	250°F
NAS 1738	NAS 1740	Al (2219, 5056) Inconel 600 (Sleeve) Monel	
		Inconel 600 Alloy Steel (Spindle) A 286	
		Monel Inconel (Lock Ring)	
NAS 1768	NAS 1740	Al (2017-T4) Monel (Sleeve)	
		Al (2017-T4) Monel (Lock Ring)	
		Al (7075-T6) Monel (Stem)	
NAS 1919	NAS 1900	Al (5056) A 286 (Sleeve) Monel	
		Al (2024) A 286 (Spindle)	
		Al (5056) A 286 (Lock Ring)	

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TABLE 213-I. Blind fasteners. - Continued

Part Number	Procurement Specification	Material	Other
Flush (100°) Head			
MS 21140	MIL-F-8975	A 286	95 ksi FSU
MS 90353	MIL-F-81177	Alloy Steel	112 ksi FSU
NAS 1399	NAS 1400	Al (5056, 2017) Monel, A 286	
NAS 1670	NAS 1675	Alloy Steel (Nut & Screw) CRES (Sleeve)	
NAS 1672	NAS 1675	A 286	1200°F (max)
NAS 1674	NAS 1675	Al (7075) (Nut) Alloy Steel (Screw) CRES (Sleeve)	250°F
NAS 1739	NAS 1740	Al (2219, 5056) Inconel 600 (Sleeve) Monel	
		Inconel 600 Alloy Steel (Spindle) A 286	
		Monel Inconel (Lock Ring)	
NAS 1769	NAS 1740	Al (2017-T4) Monel (Sleeve)	
		Al (2017-T4) Monel (Lock Ring)	
		Al (7075-T6) Monel (Stem)	
NAS 1921	NAS 1900	Al (5056) A 286 (Sleeve) Monel	
		Al (2024) A 286 (Spindle)	
		Al (5056) A 286 (Lock Ring)	

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TABLE 213-II. Blind rivets/bolts hole sizes.

Diameter Dash No.	Hole Sizes For Blind Rivets	Hole Sizes For Blind Bolts
-3	.097 - .101	
-4	.129 - .132	
-5	.160 - .164	.1645 - .1675
-6	.192 - .196	.199 - .202
-8	.256 - .261	.260 - .263
-10		.312 - .315
-12		.375 - .378
-14		.437 - .441
-16		.500 - .504

6.3 The grip increments for newly developed blind fasteners should be as listed in table 213-III.

TABLE 213-III. Grip increments.

Material Thickness to be joined (Grip)	Grip Dash Numbers
1/To - .062	-1
.063 - .125	-2
.126 - .187	-3
.188 - .250	-4

1/ See individual standards pages for minimum grip.

NOTE: Longer grip lengths are available as documented on the applicable standards page.

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6.4 Newly developed blind fasteners should be designed to be rated at one of the following shear strengths in pounds per square inch (psi):

30,000	75,000
34,000	95,000
38,000	112,000
46,000	125,000
50,000	132,000
55,000	

6.5 When using NAS 1398 and NAS 1399 fasteners, the minimum blind side sheet thickness should be as shown in table 213-IV.

TABLE 213-IV. Recommended minimum blind sheet thickness (in inches) for NAS 1398/1399.

Dia Dash No.	Rivet Shank Material	Blind Sheet Material			
		Aluminum			Titanium Ti-6Al-4V
		6061-T6	2024-T3	7075-T6	Annealed
-4	5056-F	.032	.032	.032	.016
-5	Aluminum	.032	.032	.032	.016
-6		.040	.040	.032	.016
-8		.050	.050	.032	.020
-4	2017-T4	.040	.032	.032	.016
-5	Aluminum	.040	.032	.032	.016
-6		.040	.040	.032	.016
-8		.040	.040	.032	.020
-4	Monel	.025	.025	.025	.016
-5		.032	.032	.032	.016
-6		.050	.050	.050	.016
-8		.063	.063	.063	.020
-4	A-286	.032	.025	.025	.016
-5		.032	.032	.032	.016
-6		.040	.040	.040	.025
-8		.050	.050	.050	.032

NOTE: The minimum thickness in a column may be used for other sheet materials having an equal or higher bearing ultimate strength (Fbru) value. Use of these fasteners in a thickness less than shown and in non-metallic materials shall be substantiated by test.

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6.6 By definition, an oversize/repair fastener is one specifically designed for replacement purposes. Its shank diameter limits are greater than the fastener of the same family that is intended for selection in original design. The repair fastener shank diameter provides for larger diameter holes that are due to the removal of corrosive effects, the sizing of the hole, wear, etc. The fact that a fastener's shank is different (larger) than what is normally construed to the "nominal" diameter does not require that fastener to be identified as oversize.

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CONTENTS (Continued)

SECTION 200 - STRUCTURAL FASTENER REQUIREMENTS

Requirement 201	*	24 June 1983	Bolts and Screws, Regular and Self-Locking
202	*	24 June 1983	Studs
203		24 Sept 1979	Bolts, Self-Retaining
204		24 Sept 1979	Nuts
205		12 July 1978	Inserts, Screw Thread
206	*	24 June 1983	Rivets, Solid
207			Pins, Dowel, Taper, Spring
208		24 Sept 1979	Spacers
209		12 July 1978	Washers
210		12 July 1978	Shims
211			Keys
212			Hinges
213	*	24 June 1983	Blind Fasteners
214			Structural Pin Fastening Systems - Permanent Fasteners
215		(deleted)	Panel Fasteners
216		12 July 1978	Retaining Rings
217		12 July 1978	Sandwich Constructin Fasteners
218	*		Clamps
219		12 July 1978	Control Cable and Control and Structural Rod Components; Fastening and Safetying
220		12 July 1978	Pins, Quick Operating
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222		(deleted)	Captive Screws
223			Repair/Rework Methods for Fasteners
224		(deleted)	Fuel Sealing Fasteners

SECTION 300 * NONSTRUCTURAL FASTENER REQUIREMENTS (PROPOSED)

SECTION 400 * PROPULSION SYSTEMS FASTENER REQUIREMENTS (PROPOSED)

SECTION 500 * HYDRAULIC SYSTEMS FASTENER REQUIREMENTS (PROPOSED)

SECTION 600 * ELECTRICAL/ELECTRONIC FASTENER REQUIREMENTS (PROPOSED)

SECTION 700 * SUBSYSTEMS/AUXILIARY FASTENER REQUIREMENTS (PROPOSED)

SECTION 800 * APPLICATION REQUIREMENTS (PROPOSED)

* Denotes change.
Supersedes page v of 12 July 1978

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3.1.11 Fastening system. An installed fastener, its component parts, the geometry of the hole where it affects the performance of the system, and installation and removal tooling and procedures.

3.1.12 Grip or grip accommodation. The allowable variation of material thickness in which a specific fastener can function.

3.1.13 Impedance locking. A resistance that impedes disengagement of mating fastener components.

3.1.14 Locking device. A part or mechanism designed to prevent loss of preload or disengagement of a fastener.

3.1.15 Permanent fastener. A fastener which does not require removal to service or maintain the aircraft and that normally requires the destruction of the fastener or one of its components during removal.

3.1.16 Plate nut. (See anchor nut)

3.1.17 Self locking. An attribute of a fastener or fastener assembly having an integral locking element to impede relative rotation of mating components.

* 3.1.18 Shank. The unthreaded portion of a fastener between the head and the threads, locking grooves, formed head, etcetera.

4. GENERAL STATEMENTS

4.1 Guidelines. A separate publication, "AMFRG Manual on Organization and Operations of the Aeromechanical Fastener Requirements Group for MIL-STD-1515", has been distributed to organizations participating in the preparation of MIL-STD-1515. This publication presents guidelines for the preparation and coordination of requirements for inclusion in this standard. These guidelines shall be followed for all requirement additions or changes and shall be revised as necessary by the preparing activity or his designee.

4.2 Application

a. The sections contained herein are intended to provide uniform requirements applicable to mechanical aerospace fasteners and shall be incorporated by reference in general and detail weapon systems and equipment specifications.

b. If a requirement contained herein conflicts with a requirement in the general or detail weapon system or equipment specification, the weapon system or equipment specification shall take precedence. If a requirement listed in the contents has not been published but is referenced herein, it is not applicable and the requirements of the general or detail weapon system or equipment specification shall apply.

* Denotes change.
Supersedes page dated 24 September 1979

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c. Section 100 Requirements apply only to Section 200 Structural fastener requirements (excluding subsequent proposed sections such as nonstructural, propulsion, etcetera, not published or in preparation).

4.3 Descriptive factors and symbols. The descriptive factor symbols used in this standard shall be in accordance with MS17855.

5. NOTES

5.1 Availability of Documents. Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

6. INTENDED USE AND GUIDANCE CRITERIA

6.1 Information and guidance. Paragraph 6 in all applicable requirements in this document presents information and guidance for use of that requirement or parts listed therein. The paragraph is not contractual.

6.2 Implementation. This standard may not be applicable to programs in the experimental, developmental, exploration, prototype, demonstration, validation program phases where the advantages of the standard will not be realized.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing Activity:
Air Force - 11

Reviewer Activities:
Army - AR, MI
Navy -
Air Force - 10, 99
DLA - IS

Project No. 5300-0039

* Denotes change.
Supersedes page dated 24 September 1979.

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RELEASE FOR USE OF NONAPPROVED FASTENERS

1. Purpose. Fastening systems, fasteners, and installation requirements identified and listed in the applicable requirements of this standard are approved and shall be given selection priority in new aerospace systems design. Those not listed as approved for use require a release for use by the procuring activity prior to incorporation into design, during development, or for use in assembly during production. This requirement establishes preparation and submission procedures for justification and engineering data necessary for the evaluation and use of fasteners not listed herein. It also establishes the point of contact for release requests.

2. Documents applicable to Requirement 112

MIL-D-8706	Data and Tests, Engineering, Contract Requirements for Aircraft Weapon Systems
MIL-STD-965	Parts Control System

3. Procedures for use. When a determination has been made that a fastening system or fasteners not listed herein should be used, a request for release for use, with the justification and engineering data specified herein, shall be submitted to the procuring activity in accordance with the procedures of MIL-D-8706 and MIL-STD-965. When contracts do not call out any of the above documented procedures, the requests shall be made to the contact points listed in table 112-1, through the procuring activity in accordance with procedure 1 of MIL-STD-965 and the data requirements of 3.1 of this requirement.

3.1 Engineering data required. When applicable, the following engineering data shall be furnished for each fastener not listed herein for which release for use is requested.

a. The reason an approved fastener would not be a satisfactory selection to perform the function required by the design problem or application.

b. The specific materials from which the fastener is manufactured.

c. The specific plating, coatings, surface treatments, and lubricants, as applicable. These shall be completely identified.

d. Values for the physical and mechanical properties, and available installation performance requirements, and available structural design load allowable data.

* e. An illustration of the fastener with the dimensions required to establish the geometrical material limits necessary for design selection and for determining exchangeability. This includes the geometrical material limits of installation formed fasteners after installation and clearances required for installation.

f. The maximum weight limit per one hundred of the fasteners.

g. The design and usage limitations to be applied.

h. The installation or process instruction for the part.

* Denotes change.

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3.2 Contract data requirements. Contract data requirements shall include the data specified in 3.1.

TABLE 112-I. Request contact points

Department	Submit to	Action Required
Air Force	Aeronautical Systems Division ATTN: ENFEM Wright-Patterson AFB, OH 45433 Telephone: 513-255-4158	Disapprove or Release
Army	US Army Aviation R&D Command ATTN: DRDAV-EKS P. O. Box 209 - Main Station St. Louis, MO 63166 Telephone: 314-268-5791	Disapprove or Release
Navy	Naval Air Development Center ATTN: 6013 Warminster, PA 18974 Telephone: 215-441-2833	Review and Recommendation to NAVAIR
	Naval Air Systems Command ATTN: AIR-5303 Washington, DC 20361	Disapprove or Release
All Military Services DISC/MPCAG	Defense Industrial Supply Center ATTN: DISC-EPM Philadelphia, PA 19111 Telephone: 215-697-4395	Information and Recommendation

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